Analyzing Traffic Incidents in Izmir

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***Abstract*—Traffic incidents in Izmir impose severe life-threatening risks and substantial economic costs on citizens and municipal authorities. Although this problem is prevalent, comprehensive data-driven analyses of the city's incident patterns remain scarce, limiting evidence-based intervention planning. This study addresses this gap by analyzing 21,161 traffic incident records from Izmir's metropolitan area spanning December 2021 to September 2025, applying classification, time-series regression, clustering, and ensemble learning methods to identify spatiotemporal patterns and develop predictive models for incident severity and daily counts. Analysis reveals extreme spatial concentration: Konak district accounts for more incidents than all other districts combined, with critical infrastructure hotspots, particularly three bridge overpasses, responsible for 13% of recorded incidents. Temporal analysis demonstrates systematic severity variation: late-night incidents exhibit 88% injury/fatal proportions compared to 22-30% during peak hours, indicating dramatically elevated injury risk at night. Predictive models achieved 74% accuracy in severity classification and highly accurate monthly forecasts (7% error). Unsupervised clustering categorized locations into distinct risk profiles based on incident volume, severity, and intervention times, confirming Konak as a unique high-intensity outlier. These findings enable targeted infrastructure improvements at hotspots, time-specific enforcement strategies, and risk-tiered resource allocation based on quantitative evidence.**

Our code for this project is available at [1].

I. Introduction

Traffic incidents are a prevalent problem in Izmir, imposing substantial costs on both citizens and the municipality, including life-threatening risks, economic losses, and operational burdens. Providing concise, data-driven evidence can accelerate decision-making and lead to precautionary actions that lower incident rates. While national studies exist, recent work centered on Izmir's incident patterns was scarce. This study analyzes a dataset of traffic incidents from the Izmir metropolitan area.

In this study, the incident records from the Izmir Municipality Open Data Portal were analyzed, including timestamps, location data and incident information. The goal was to determine traffic incident patterns, summarizing locations and times at which risk concentrates across the metropolitan area.

[1]https://github.com/HuseyinYontar/Analyzing_Traffic_Incidents_in_Izmir

Fig. 1. Map of the Izmir metropolitan area.

II. Background and Related Work

Numerous studies have investigated the spatial and temporal relationships of traffic incident occurrence using various analytical and data-driven methods.

In a study conducted by Cakıcı on Izmir [1], it was shown that the highest number of accidents in the years between 2020-2023 occurred in the autumn and summer seasons. In addition, least number of traffic accidents occurred during 2020 spring season due to the COVID-19 pandemic.

Another study conducted by Haybat and Karakas [2] reveals that incidents were focused on the high mobility regions of Izmir, especially in the south of Karsıyaka district, all over district of Konak, northeast of Karabaglar district, and northwest of Buca. Furthermore, the study reveals that while the number of incidents increased between 2010 and 2014, the total coverage area declined, suggesting a higher spatial density of incidents in later years.

According to a study performed by Kumar and Toshniwal [3] on road accidents in Gujarat, the time periods between 20:00 to 22:00 and 22:00 to 00:00 were identified as the most accident-prone invervals when the 24-hour day was divided into two-hour periods.

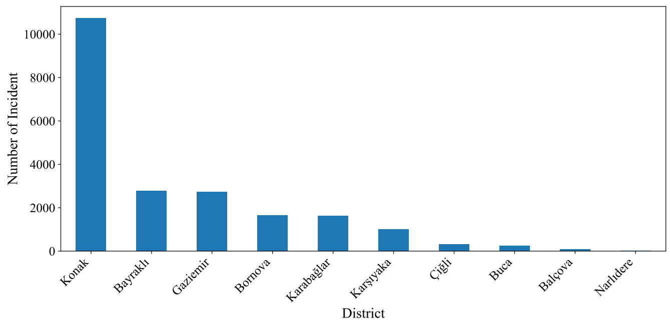

Fig. 2. Number of incidents per district between 2021-12-01 and 2025-09-28.

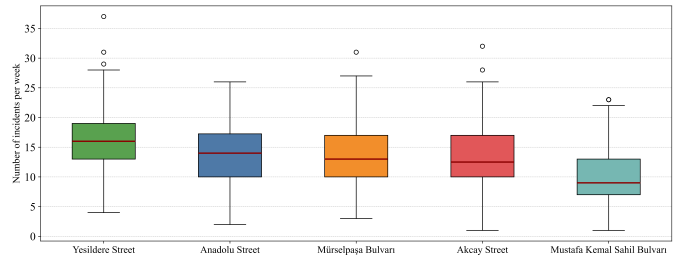

Fig. 3. Box-plot showing weekly incident counts for top 5 most incident-dense streets in İzmir

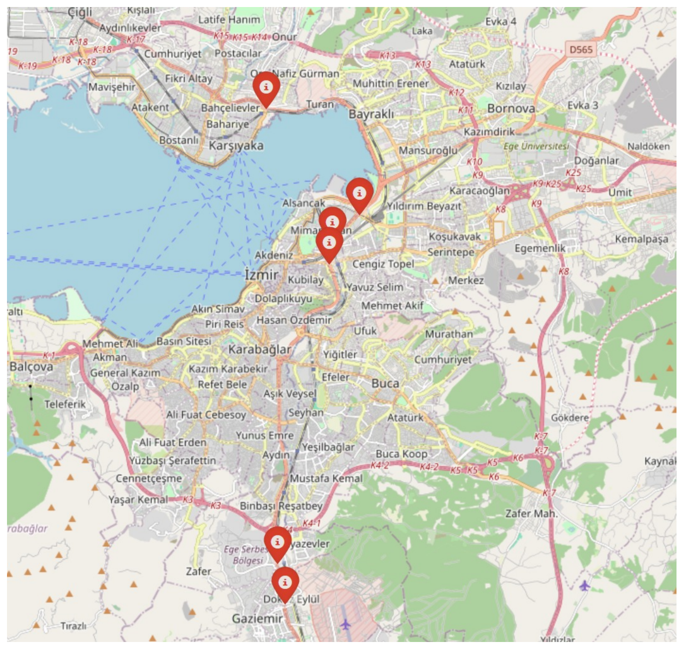

Fig. 4. Demonstrating Hotspot Incident Points of Izmir

Cardelino (1998) [4] analyzes hourly traffic-counter data and reports the familiar weekday rush-hour double peak, contrasted with a relatively flatter weekend profile. Although framed around emissions, the study highlights the value of distinguishing traffic pattern types in time-of-day analysis.

Retallack and Ostendorf [5] report that crash frequency increases non-linearly as congestion rises. Their models, estimated on over five million hourly observations from 120 intersections, show that higher congestion was associated with disproportionately larger gains in expected crash counts, especially at the most congested levels. In short, the results indicate a positive, non-linear correlation between traffic congestion and crash rates.

III. Data Set Description

The data set used in this study was provided by the Open Data Portal of Izmir Metropolitan Municipality and authored by Izmir Transportation Center Directorate (IZUM) [6]. The data set includes detailed records of vehicle breakdowns and traffic accidents that occurred within the metropolitan area of Izmir. The metropolitan area of Izmir comprises the central districts of Çiğli, Karşıyaka, Bayraklı, Bornova, Konak, Buca, Karabağlar, Gaziemir, Balçova and Narlıdere, as illustrated in Figure 1.

The data set was accessed on October 11, 2025, the period between December 1, 2021, and September 28, 2025 was covered by the observations. The raw data set consisted of 21161 samples and 7 attributes. Each sample contained the street name, the direction of travel with 16 missing values, the location description with 14 missing values, the type of incident as nominal data type and the date of the incident, the accident time, and the intervention time with 5193 missing values as interval data type.

The descriptive information regarding the data set including mode and median values can be found in tabular form in Table I.

IV. Preprocessing

As part of the preprocessing stage, the data set was examined to understand spatiotemporal incident patterns across Izmir's metropolitan districts and its most incident-dense streets.

Figure 2 demonstrates the total number of incidents by metropolitan districts. The bar chart shows that the highest number of incidents occurred in Konak, followed by Bayraklı and Gaziemir. Bornova, Karabağlar, and Karşıyaka also contribute considerably to the incident counts. In contrast, Çiğli, Buca, and Balçova contribute only a small share. The bar chart reveals that the total number of incidents in Konak was higher than the combined total of all other districts, indicating that Konak was the most incident-dense district in the Izmir metropolitan area.

Figure 3 illustrates the box plot of the weekly incident counts for the five most incident-dense streets of Izmir. From the box plot, Yesildere Street has the highest median weekly incident count, indicating a higher typical number of incidents than the other streets. Additionally, both Akçay Street and Mustafa Kemal Sahil Boulevard display lower minimum weekly incident counts compared to the other streets. Another notable pattern was that outliers were observed only on the upper end of the distributions, with no lower-end outliers present, indicating a right-skewed pattern in weekly incident counts.

TABLE I
DESCRIPTION OF EACH FEATURE IN THE RAW DATA SET.

Attribute	Type	Median	Mode	Number of Distinct Values	Number of Missing Values	Minimum	Maximum
Date	Interval	2023-07-18	2022-04-29	1397	0	2021-12-01	2025-09-28
Street	Nominal	-	Yeşildere Street	183	0	-	-
Direction	Nominal	-	City Center	157	16	-	-
Location	Nominal	-	Hilal Bridge Overpass	2032	14	-	-
Incident Type	Nominal	-	Property Damage Only	30	0	-	-
Incident Time	Interval	15:30	17:29	1344	0	00:00	23:59
Intervention Time	Interval	15:52	18:10	1295	5193	00:00	23:59

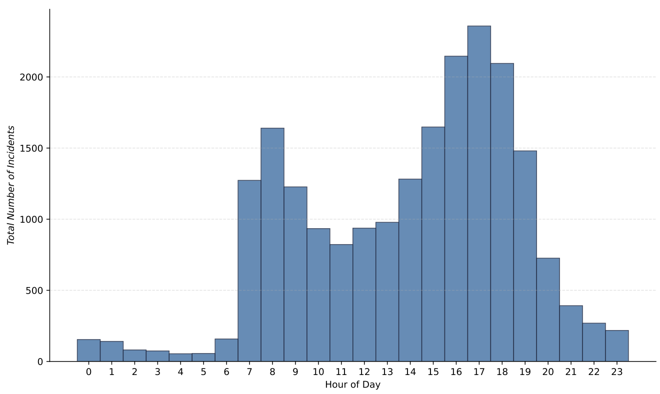

Fig. 5. Hourly incidents histogram

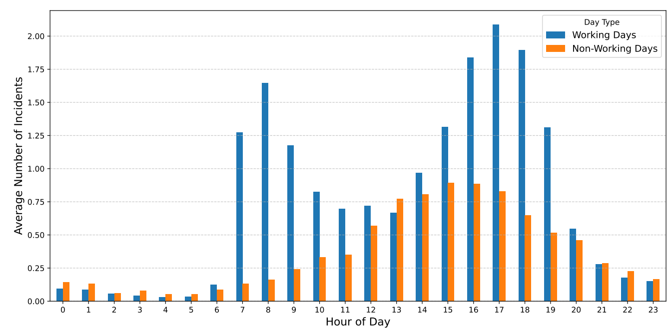

Fig. 6. Hourly incidents bar chart comparing working days and non-working days.

Figure 4 displays the most incident-dense locations in Izmir: Mürselpaşa Boulevard on Hilal Bridge overpass with 954 incidents, Yeşildere Avenue on Tepecik Bridge overpass with 949 incidents, Mürselpaşa Boulevard on DGM Bridge overpass with 886 incidents, Akçay Avenue inside the Free Zone underpass with 521 incidents, Akçay Avenue inside the Gaziemir underpass with 497 incidents, and Anadolu Avenue after Naldöken Bridge with 484 incidents. This figure was crucial for pinpointing hotspots, where focused interventions produce the greatest impact per unit of effort and cost.

Figure 5 illustrates the distribution of incident counts across the hours of the day. The number of incidents exhibits an upward trend during the early morning hours, reaching its first peak at around 08:00. After this point, incident occurrences decline until midday. Following noon, incident counts rise again, with the highest rates observed around 17:00, which aligns with the end of typical working hours. Incident counts then decline sharply throughout the evening and continue to decrease into the late-night period.

Figure 6 compares the average hourly incident counts on working and non-working days. On working days, incidents begin to increase after 07:00, reach a morning high at 08:00, and then decline toward 11:00. Thereafter, average incident count gradually rise again and reaches its peak at 17:00. Following this point, incidents decline sharply until around 20:00 and continue decreasing into the late hours.

On non-working days, including weekends and national holidays, total incident numbers rise steadily through the morning, peak around 16:00, and then gradually decline into the late-night hours. These findings on İzmir's traffic patterns, particularly the pronounced mid-afternoon peak on non-working days, are consistent with Cardelino's weekend traffic volume distribution results [4].

By examining hourly incident patterns, weekdays exhibit a double peak that aligns with established weekday congestion signatures, while weekends show a flatter profile with a pronounced mid-afternoon high [4]. Retallack and Ostendorf [5] further show that expected crash counts rise non-linearly as congestion intensifies. According to their analysis, heavy traffic yields larger marginal increases in expected crashes, indicating a positive, non-linear congestion and crash rates.

Figure 7 shows how the distribution of accident types varied by hour of day, using a stacked bar chart of hourly totals with within-hour percentage shares. In this chart, injury and fatal incidents were combined into a single injury/fatal category, while property-damage-only crashes remain as a separate group due to their high frequency. Less common incident types such as pile-up and similar categories were excluded from the figure, but this omission was not expected to affect the overall pattern due to their low occurrence. Although the number of accidents per hour decreased after 22:00, the analysis indicated that the ratio of injury or fatal crashes to property-damage-only incidents increased until 06:00. This figure suggests that while daytime traffic density contributes to high amount of incidents, nighttime incidents were observed to have smaller amount of incidents with higher injury and fatal share.

Figure 8 presents two bar charts: the top panel shows the number of distinct streets by district, and the bottom panel shows the average number of incidents per street. The first inference concerns Konak, across both panels, Konak ranks first in both street count and average incidents per street.

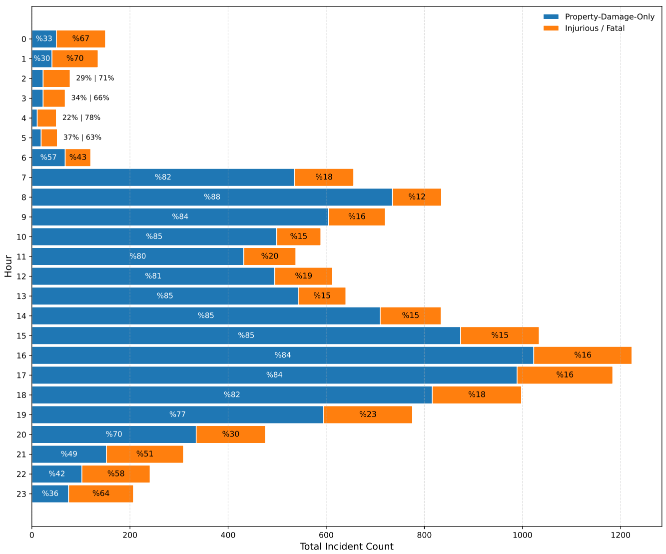

Fig. 7. Distribution of total accidents per hour including their accident type percentages

Because overall incident totals were influenced by both the number of streets and the per-street average, Konak's leading total in Figure 2 was consistent with these results.

Another inference concerns Bayraklı. Figure 2 shows Bayraklı has the second-highest total incident count. In Figure 8, Bayraklı ranked sixth in street count and second in average incident count per street. This combination indicated very high incident density in Bayraklı, compensating for its smaller number of streets relative to other districts.

Finally, Buca ranks second in the number of distinct streets but eighth in average incidents per street. A similar pattern appears in Balçova, Narlıdere, and Çiğli as well, indicating lower incident density despite many distinct streets. This make these districts relatively less important when prioritizing preventive interventions.

Beyond street counts and incidents per street, these patterns may also reflect underlying exposure and built-environment differences between districts. For example, central districts such as Konak typically have higher population and employment density, more through-traffic, and a denser road network, all of which increase opportunities for crashes. Likewise, a district with relatively few streets but a high incidents-per-street value (such as Bayraklı) could indicate concentrated traffic on specific arterials or local design and control issues at key junctions, rather than simply a larger number of roads.

Figure 9 illustrates the relationship between the monthly number of distinct streets and total incidents, along with their ratio, computed by dividing the total number of incidents by the number of distinct streets each month. The top panel combines a bar chart of total monthly incidents and a bar chart of the monthly count of distinct streets, where both gradually decline across the period. The bottom panel plots the ratio of total incidents to distinct streets. This series remains fairly steady, typically between 10 and 16, which aligns with the

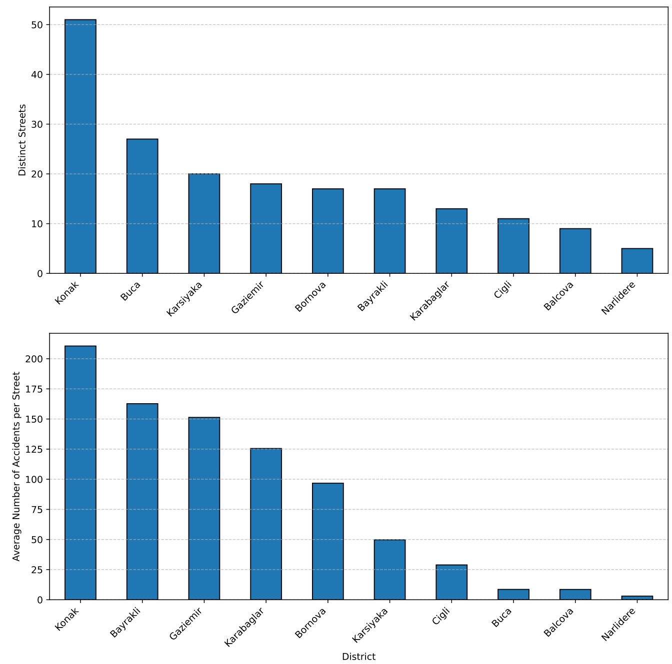

Fig. 8. Number of streets (top) and average accidents per street (bottom), by district

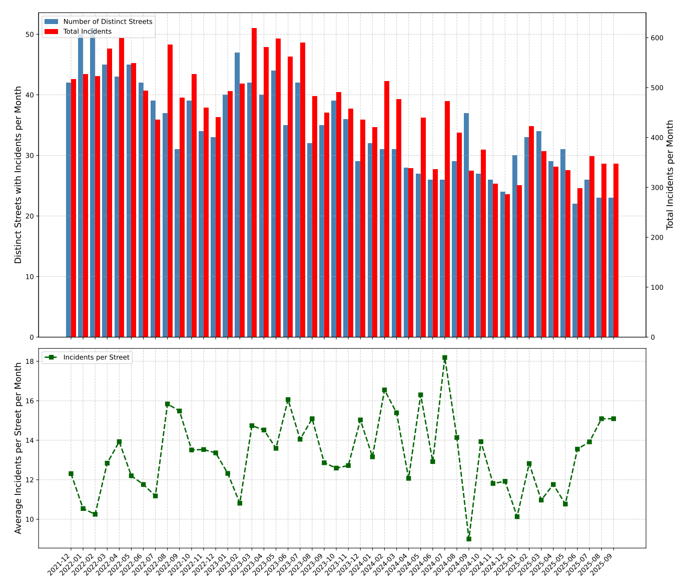

Fig. 9. Monthly incident summary for İzmir, showing total incidents and the number of distinct streets with incidents (at top, combined together), and the average number of incidents per street (at bottom), computed as total incidents divided by distinct incident-involved streets.

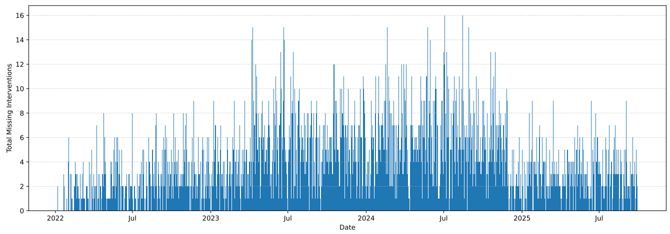

Fig. 10. Bar chart indicating the number of missing intervention time values in timeline

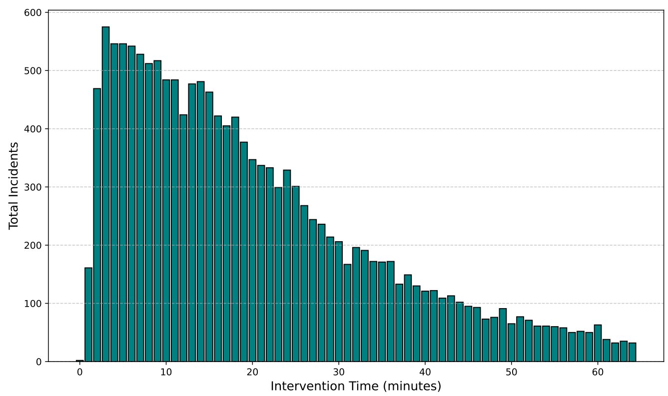

Fig. 11. Demonstrating total incidents in different intervention times

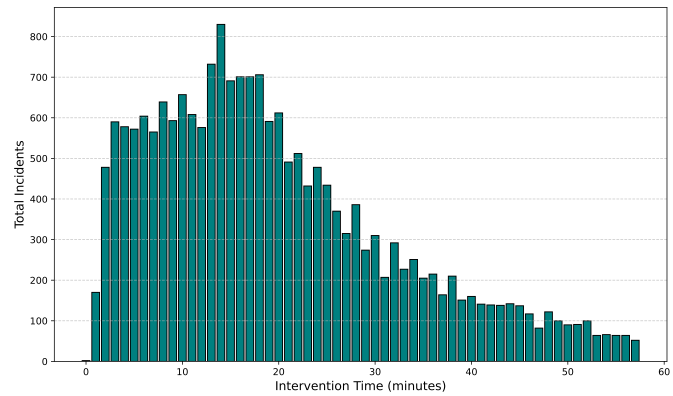

Fig. 12. Illustrating total incidents in different intervention times after imputing

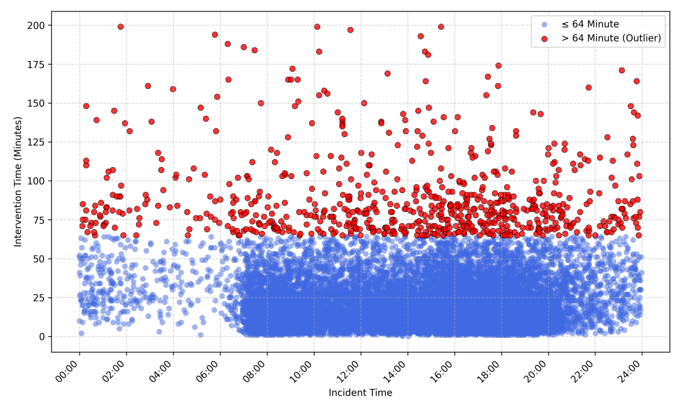

Fig. 13. Scatterplot Indicating Incident Time and Intervention Time Relations

gradual decrease trend shown in the top and middle panels. In summation, the decrease in total incidents appears to align with a decrease in the number of distinct streets involved.

Figure 10 shows how many entries in the intervention time column were missing each day. There were in total 5193 missing entries in the raw data set. On inspection, the figure shows entries for each day, indicating that missing entries were uniformly distributed over the timeline.

Figure 11 shows the distribution of the duration between accident time and intervention time. The figure follows a right-skewed distribution, indicating that incidents with short intervention times were more common than those with long intervention times.

After examining the patterns in the missing intervention time entries, it was observed that there were no time periods where data was missing in bar blocks. In conclusion, it was decided to impute these missing intervention time values. To achieve this goal, first of all, the median duration between accident time and intervention time was calculated within detailed groups defined by accident type, location, time interval, street and working day status. Then, missing values were imputed. In the Figure 12, the distribution of intervention time with imputed values can be seen.

Figure 13 plots incident occurrence time against intervention time in minutes. Blue points show points with intervention times under 64 minutes and red points mark outliers with intervention times above 64 minutes. The 64 minute cutoff was derived from the intervention-time distribution using the standard boxplot rule: the interquartile range was 22 minutes, yielding an upper outlier threshold of 64 minutes and a lower bound of -24 minutes, which was truncated to 0 since negative values were not meaningful. In addition, 16 records with intervention times exceeding 200 minutes were treated as extreme anomalies and discarded from the analysis to prevent distortion of the computed statistics. The scatterplot suggests that during high-traffic hours the outlier rate was lower, whereas during late-night and early-morning periods which were outside of regular working hours, display a higher outlier proportion, which could be explained by lack of available staff that could concern incidents.

Figure 14 provides heatmap, demonstrating rate of correlation between selected attributes. To construct the heatmap, we first selected the rows with the frequent incident types: vehicle breakdown, injury/fatal accident and property damage only accident, then partitioned the timeline into intervals with approximately equal incident counts. This procedure resulted in six time intervals, each containing approximately 3,400 incidents. The first interval covers the late evening to early morning period (20:00–07:54). The second interval spans the early morning hours (07:55–10:32). The third interval corresponds to the late morning and early afternoon (10:33–14:23). The fourth interval represents the mid-

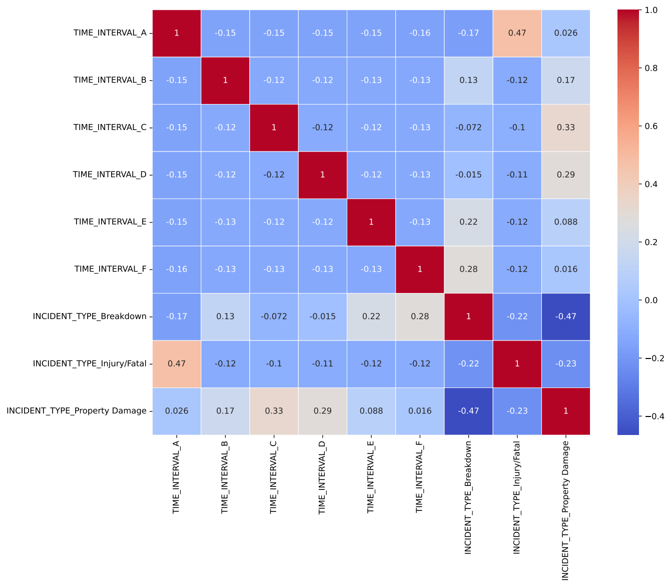

Fig. 14. Correlation heatmap of time intervals and several incident types

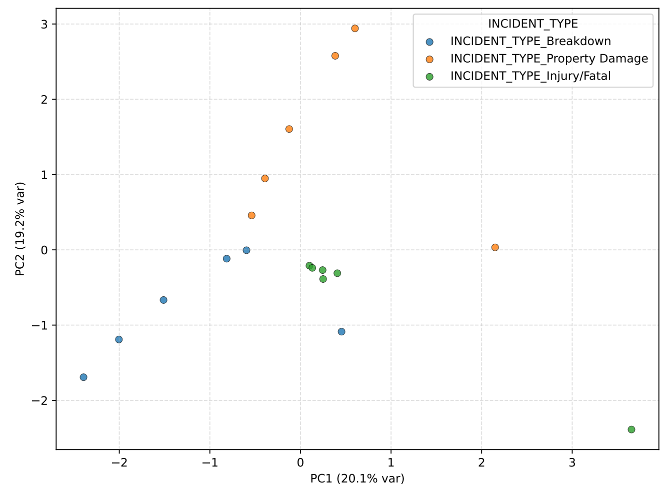

Fig. 15. Principal Component Analysis (PCA) between different incident types

afternoon period (14:24–16:30). The fifth interval includes the late afternoon and early evening (16:31–18:01). Finally, the sixth interval covers the early evening period (18:02–19:59).

Indicator columns were then created to distinguish between the most frequent incident categories: breakdown-related events, injury or fatality cases, and property-damage-only incidents. Next, normalization process was applied to achieve reasonable results. Because these features were nominal, we applied one-hot encoding to obtain numeric representations, followed by normalization to conduct healthy comparisons across variables. Subsequently, encoded variables were multiplied with their respective total incident numbers.

The heatmap reveals several interesting patterns. Injury or fatality-related incidents shows a relatively high positive correlation number of 0.47 with the earliest time interval. Since that interval comprises 20:00 to 07:54 hours, the injury incident rates were high as supported in Figure 7. In addition, property-damage-only events were more common during the mid-morning to late-afternoon period (approximately 07:55–16:30), with correlation values reaching up to 0.33. Finally, breakdown-related incidents display higher occurrence rates in the late-afternoon and evening hours (16:31–19:59), showing correlations of up to 0.28.

After discussing the figures, some additional findings about the attributes were discussed. In the GitHub repository of this project[2] and in Figure 14, there was a correlation matrix of some one-hot encoded categorical features. The purpose of this analysis was to identify potential relationships between different time intervals and incident types.

After searching for correlations between the attributes, in order to predict the type of the incidents, Random Forest Classifier and Decision Tree Classifier with Sequential Feature Selector were used for attribute selection. It was not ended with a significant result when the selection was done with Random Forest Classifier. For the Decision Tree Classifier, the best result was founded in the following columns: Street, Direction, District, Season, isHoliday and Time Interval. The optimal subset was reached after 27 features with a best cross-validation accuracy of 0.5181. The result can be founded in the GitHub repository[3].

In the project's GitHub repository[4], Principal Component Analysis (PCA) were conducted. First, data-cleaning procedures were applied. Then, records with missing values were removed. Next, one-hot encoding were performed to convert categorical variables to numeric form. To examine alternative correlation structures, PCA were run on both the original Excel dataset and an edited version that included derived attributes such as intervention time, day type, and season. The best resulting PCA yielded with the dataset used in the correlation matrix that contain incident types which can be seen in Figure 15. In the matrix, a clear clustering of injury and fatal crashes can be observed.

V. Classification / Regression

For the classification task, the objective was to predict whether a future incident would be life-threatening or non-life-threatening. To achieve this goal, several supervised learning algorithms were employed, including Random Forest, k-Nearest Neighbors (KNN), Multilayer Perceptron (MLP) neural network, Logistic Regression, and Decision Tree. These methodologies demanded preparation of the dataset to make it suitable for classification tasks. Firstly, some types of recorded

[2]https://github.com/HuseyinYontar/Analyzing_Traffic_Incidents_in_Izmir/blob/main/heatmap/onehot_heatmap.pdf

[3]https://github.com/HuseyinYontar/Analyzing_Traffic_Incidents_in_Izmir/blob/main/attribute_selection/sfs_decision_tree_accuracy_curve_without_cadde.pdf

[4]https://github.com/HuseyinYontar/Analyzing_Traffic_Incidents_in_Izmir/tree/main/pca

instances were removed to reach a simpler dataset that was clean from inconsistent or invalid incident type records. This goal was achieved by selecting four high-frequency categories in the raw data set and removing records that belong to other incident types: injury, fatal, property-damage-only, and vehicle breakdown. Then, injury and fatal crashes were merged into a single injury/fatal category, representing life-threatening incidents, while property-damage-only crashes and breakdowns were grouped as non-life-threatening events. As a result, the target variable was a binary indicator of incident criticality, enabling authorities to prioritize preventive measures according to the anticipated severity of future incidents.

In addition to the preprocessing applications, remaining records were augmented by using the incident timestamp. As a result of this step, the day of week, month, day type (weekday, weekend, or public holiday), and working day columns were added as new separate columns. Moreover, the incident hours were encoded using two complementary approaches. First, the incident time was discretized into hourly intervals, represented as [0-1), [1-2), . . . , [22-23), [23-24). Secondly, the interval-based discretization introduced in the heatmap analysis was incorporated as an additional attribute. Furthermore, seasonal information about the incidents was added. Finally, to capture spatial structure of the districts, street and location information were used to detect the district attribute manually.

In the classification analysis, only a small portion of the data set was used, consisting of 5554 incidents. The goal behind this choice was to mitigate class imbalance by applying a simple undersampling scheme. This method was motivated by the fact that life-threatening incidents consist of a small portion of the whole set. Accordingly, all life-threatening incidents were retained, and an equal number of non-life-threatening incidents were randomly sampled to form a balanced data set. The training portion of the data set was composed of 2160 life-threatening and non-life-threatening incidents which holds approximately 77% of the selected data. The remaining incidents were used in test set with 617 incidents per class, forming a group of 1234 observations in total which holds around 23% of the data set. After all, applying the undersampling procedure helped the model by reducing the risk of the classifiers being dominated by the majority class.

Table II shows the performance assessment of the classification models using overall accuracy, F1-score, class-wise precision and recall, as well as the confusion-matrix components (TN, FP, FN, TP).

A grid search was conducted to optimize hyperparameters for each machine learning model separately. For the neural network, the grid search using PyTorch-based neural network implementation resulted in the selection of a multilayer perceptron (MLP) architecture. The MLP architecture achieved the highest accuracy of 0.7382, with the following precision and recall values for both classes: life-threatening (precision: 0.79, recall: 0.65, F1-score: 0.71) and non-life-threatening (precision: 0.70, recall: 0.82, F1-score: 0.76). Logistic Regression obtained a comparable accuracy of 0.7253, followed by Random Forest with 0.7083 accuracy.

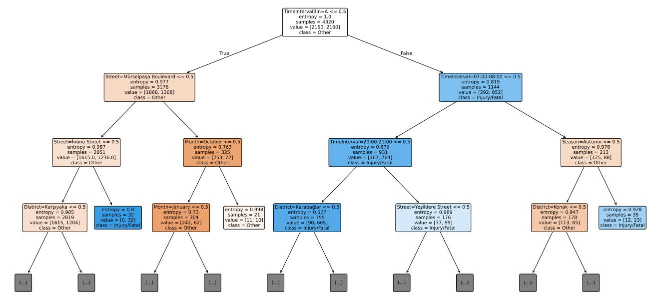

Fig. 16. Decision Tree model, indicating the severity of the incidents

Another model trained in the classification analysis was a decision tree which was shown in Figure 16. When the decision tree model was examined, the most important attribute was found to be time interval, covering the late evening to early morning period (20:00–07:54). Full version of the decision tree can be founded in the Github repository[5].

Furthermore, the Random Forest, Logistic Regression, and KNN models were also evaluated within the classification analysis. The KNN classifier achieved an accuracy of 0.6515; its hyperparameters were tuned via Grid Search, with the best configuration using 11 neighbors. The Random Forest model reached an accuracy of 0.7212 and benefited from Grid Search, obtaining its optimal setting with 400 base estimators. Finally, Logistic Regression achieved an accuracy of 0.7252, with its parameters similarly selected using Grid Search.

In addition to the classification analysis, several regression methods were employed to predict monthly total incident counts using supervised learning. To prepare the data set for regression models, a univariate time series of daily incidents was created by applying a discretization method which involves grouping incidents by calendar dates and then counting the number of events per day. Daily counts were then aggregated to obtain monthly total incident counts, which were modelled as a continuous response variable using several supervised learning methods, including conventional regression algorithms and a neural-network model.

To capture systematic differences between working days and holidays, a day-type feature was defined to indicate whether a given date was a weekday, weekend, or special day (public holiday), a seasonality feature indicating whether the observation belongs to winter, spring, summer, or autumn, and a month-name feature indicating the calendar month of the observation. These nominal variables were then transformed into a set of binary indicator features via one-hot encoding, allowing the models to exploit these patterns, which resulted in an improvement in the predictive accuracy of the regression models that was validated by the decrease in mean absolute percentage error.

[5]https://github.com/HuseyinYontar/Analyzing_Traffic_Incidents_in_Izmir/blob/main/classification/decision_tree_for_different_types/decision_tree_full.pdf

TABLE II
DETAILED PERFORMANCE METRICS OF THE CLASSIFICATION MODELS TRAINED

Model	Accuracy	Precision		Recall		F1-score		Confusion Matrix			
		Life-threatening	Non-life-threatening	Life-threatening	Non-life-threatening	Life-threatening	Non-life-threatening	TN	FP	FN	TP
Random Forest	0.7212	0.75	0.70	0.66	0.79	0.70	0.74	485	132	212	405
KNN	0.6515	0.71	0.62	0.52	0.78	0.60	0.69	484	133	297	320
MLP (NN)	0.7382	0.79	0.70	0.65	0.82	0.71	0.76	507	110	213	404
Logistic Regression	0.7253	0.78	0.69	0.63	0.82	0.70	0.75	506	111	228	389
Decision Tree	0.7050	0.75	0.67	0.61	0.80	0.67	0.73	495	122	242	375

The models were trained using the data starting from 23 December 2021 until the end of January 2025. The training period covers 1,136 days, which constitutes approximately 85% of the data. The remaining data between February and August 2025 were selected as the test set, covering 212 days, which corresponds to approximately 15% of the instances. This split was chosen so that the models were trained on a long historical period while being evaluated on a more recent interval with different seasonal patterns.

The regression analysis on this prepared data was conducted by using multiple models. Two of these were MLP Neural Network and Linear Regression. The third model was selected among different time series models: Auto-regressive (AR), Moving Average (MA), Exponential Smoothing, Auto-regressive Moving Average (ARMA), Auto-regressive Integrated Moving Average (ARIMA) and Seasonal Auto-regressive Moving Average (SARIMA); the model with the lowest MAPE was chosen as the final time-series model. All models were trained on the same training set.

During evaluation, all models were tested on the test set using a day-by-day prediction approach. For each day t in the test period, the input for the models was constructed by using the actual observations from the previous 21 days ($t-21,\ldots,t-1$), and a one-day-ahead forecast for day t was generated. This procedure was repeated sequentially for every day in the test set.

To obtain performance metrics, the daily prediction errors were aggregated on a monthly basis. For each month in the test period, the Mean Absolute Percentage Error (MAPE) and Root Mean Squared Error (RMSE) were computed using the daily observed and predicted values within that month. These monthly error measures were subsequently used to compare the performance of the trained models for different streets and districts which can be observed in Table III.

For visualizing the findings of the regression models, several figures were plotted. Figures 17–21 illustrate out-of-sample predictions for the period February–August 2025 at five high-priority locations: Konak district, Gaziemir district, Anadolu Street, Yeşildere Street, and Mürselpaşa Boulevard. In Konak (Figure 17), the MLP model achieves accurate predictions, yielding a test MAPE of 7.20% and RMSE of 15.86, while the SARIMA specification attains a slightly lower MAPE (6.60%) but higher RMSE (18.50); linear regression performs noticeably worse, with 10.71% MAPE and an RMSE of 24.28. In Gaziemir (Figure 18), the Auto-ARIMA time-series model provides the lowest errors (10.75% MAPE, RMSE 6.94), the MLP model yields moderate errors (20.01% MAPE,

TABLE III
SUMMARY OF MONTHLY MAPE AND RMSE VALUES OF REGRESSION MODELS FOR TESTED LOCATIONS (FEBRUARY 2025 – AUGUST 2025)

Location	Neural network (MLP)		Linear regression		Time-series model*	
	MAPE (%)	RMSE	MAPE (%)	RMSE	MAPE (%)	RMSE
Konak	7.20	15.86	10.71	24.28	6.60	18.50
Gaziemir	20.01	9.77	30.14	14.07	10.75	6.94
Yeşildere Caddesi	17.56	8.54	34.07	15.85	22.65	13.31
Mürselpaşa Bulvarı	6.19	3.68	18.82	10.19	14.35	11.23
Anadolu Caddesi	12.20	7.83	11.27	7.68	8.90	5.71

*Konak: SARIMA(1,1,1,7); Gaziemir: ARIMA(0,1,1); Yeşildere Street: SARIMA(1, 1, 1, 7); Mürselpaşa Boulevard: ARIMA(2, 1, 2); Anadolu Street: ARIMA(2, 1, 2).

RMSE 9.77), and linear regression again produces the largest errors (30.14% MAPE, RMSE 14.07). On Anadolu Street (Figure 19), only the ARIMA model achieves the lowest error rates, outperforming the MLP and linear regression models in terms of accuracy (8.90% MAPE, RMSE 5.71, compared with 12.20% MAPE, RMSE 7.83 for the MLP and 11.27% MAPE, RMSE 7.68 for linear regression). For Yeşildere Street (Figure 20), error levels were higher for all models, with test MAPE scores ranging from 17.56% to 34.07% and RMSE values between 8.54 and 15.85, where the MLP again performs best and linear regression worst. On Mürselpaşa Boulevard (Figure 21), the MLP provides the best fit with 6.19% MAPE and RMSE of 3.68, followed by the ARIMA model (14.35% MAPE, RMSE 11.23), and lastly linear regression with 18.82% MAPE and RMSE of 10.19. These location-specific plots indicate that neural networks and selected time-series models perform better at predicting monthly incidents compared to linear regression models. Figure 22 and Figure 23 summarize the overall comparison of these models by showing the mean absolute percentage error and root mean squared error of the three approaches across all five locations. The bar charts reveal comparatively high MAPE and RMSE values for the linear regression model, supporting the prior findings. Yeşildere Street stands out as the most difficult location to predict, with all three models exhibiting relatively high MAPE and RMSE values, while Mürselpaşa Boulevard and Anadolu Street shows small error rates overall. These patterns indicate non-linear and time-series models were better suited than a simple linear specification, and that model choice should be tailored to individual locations rather than relying on a single global model.

VI. ASSOCIATION MINING

Following the use of machine learning models to predict incident patterns, several association mining methodologies

Fig. 17. Comparison of observed and predicted monthly traffic incident counts in Konak between February 2025 and August 2025. The panels show (top) a multilayer perceptron neural network, (middle) a linear regression model, and (bottom) the best-performing SARIMA specification, with model predictions plotted against the observed monthly incident counts.

Fig. 18. Comparison of real and predicted monthly traffic incident numbers in Gaziemir between February 2025 and August 2025. The panels show (top) a multilayer perceptron neural network, (middle) a linear regression model, and (bottom) an ARIMA time series model specification, with model predictions plotted against the observed monthly incident counts.

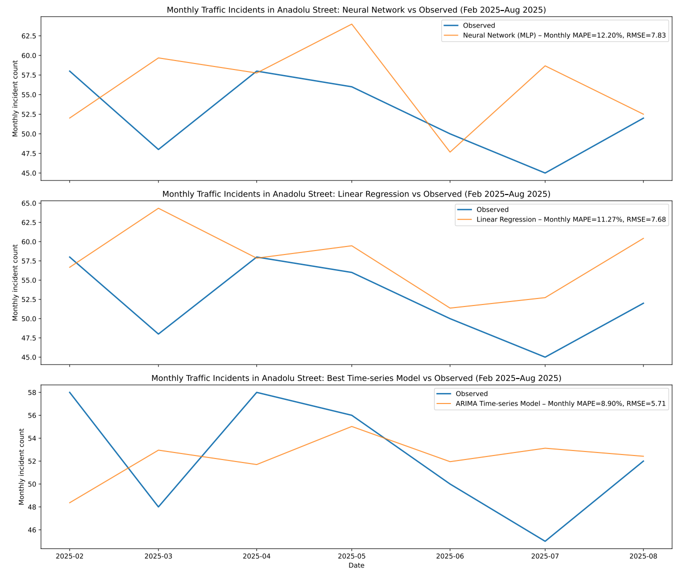

Fig. 19. Side-by-side comparison of actual and model-predicted monthly numbers of traffic incidents on Anadolu Street between February 2025 and August 2025. The panels display (top) a multilayer perceptron neural network, (middle) a linear regression model, and (bottom) an ARIMA model, with each model's predictions overlaid on the observed monthly incident counts.

Fig. 20. Monthly traffic incident totals in Yeşildere Street: observed values versus regression model estimates between February 2025 and August 2025. Shown in the panels were, from top to bottom, a multilayer perceptron neural network model, a linear regression model, and a SARIMA time series model, each with its monthly incident predictions plotted against the observed values.

Fig. 21. Comparison between observed and forecasted monthly traffic incident counts on Mürselpaşa Boulevard between February 2025 and August 2025. In the panels, the top plot corresponds to a multilayer perceptron neural network, the middle to a linear regression model, and the bottom to an ARIMA time series model, with predicted monthly incidents plotted with on the observed data.

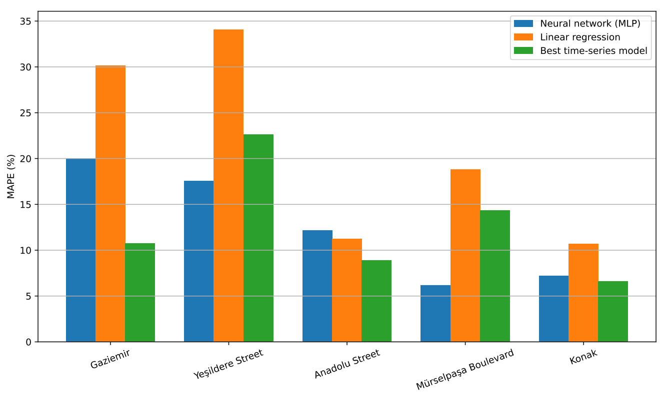

Fig. 22. Mean absolute percentage error (MAPE) of neural network, linear regression, and best time-series models for monthly incident predictions on selected streets and districts.

were considered. Specifically, the Apriori algorithm was employed to gain further insights from the dataset by identifying recurring attribute patterns.

For the Apriori algorithm two different data sets were used. The first data set consists of 20242 rows, like the original raw data set with a few added columns. As for the columns, day information which tells which day of the week the incident occurred, and month column were added. The second data set was using aggregated daily incident counts, consisting of 1398 rows. For both data sets a column indicating equal width interval bins were added. The first set had the time interval column that was also used for the heatmap, the second data set had its unique bins, consisting of different daily incident counts starting from [0, 7.8] till (31.2 ,39]. For trying different

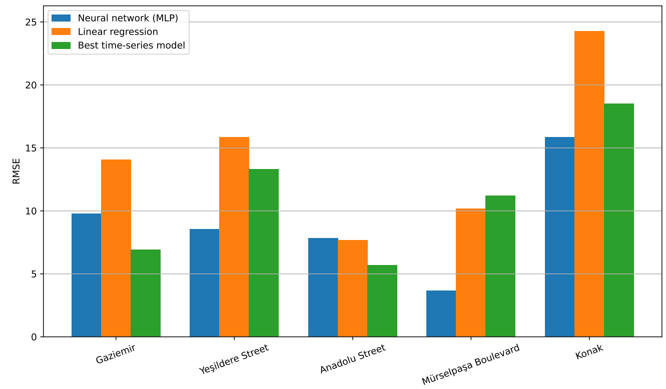

Fig. 23. Root mean square error (RMSE) of neural network, linear regression, and best time-series models for monthly incident predictions on selected streets and districts.

TABLE IV
IDENTIFIED ASSOCIATION RULES FOR THE FIRST DATA SET.

Antecedents	**Consequents**	**Support**	**Confidence**	**Lift**
Weekday	Konak	0.409	0.501	0.981
Weekday	Property Damage	0.382	0.468	0.971
Weekday	Vehicle Breakdown	0.337	0.413	1.083
Konak	Property Damage	0.240	0.469	0.974
Weekday	Spring	0.226	0.276	0.979

data sets time columns gave more meaningful results than using district columns, especially for the first data set, due to low frequency of the daily incident counts in some districts or locations.

Consequently, the relationships between variables were evaluated using support, confidence, and lift metrics. In the first dataset, one observation indicated that if an incident occurred on a Weekday, it was likely to be in Konak, yielding a support value of 0.409, a confidence value of 0.501, and a lift value of 0.981. However, the Apriori algorithm results appear to lack significance due to the dominance of specific values within the attributes. For example, Konak was the district with the highest incident frequency, and weekdays occur significantly more often than weekends. The values obtained from the algorithm follow a general pattern of lift values close to 1, indicating the results were not meaningful. The top five association rules, ranked by their support values along with their corresponding metrics, were presented in Table IV.

For the second data set, aggregated daily incident counts were utilized, yielding the top 5 results of the Apriori algorithm based on support values. These results, which include support, confidence, and lift values, were presented in Table V. A distinct pattern from the table was the association between Sunday and the [0, 7.8] incident count interval, characterized by a strong lift value of 3.588. This demonstrates a strong predictive relationship, suggesting that Sundays were significantly more likely to have low incident counts compared to other days.

TABLE V
IDENTIFIED ASSOCIATION RULES FOR THE SECOND DATA SET.

Antecedents	Consequents	Support	Confidence	Lift
Winter	Incident Count = (7.8, 15.6]	0.102	0.393	1.038
Autumn	Incident Count = (7.8, 15.6]	0.097	0.452	1.192
Sunday	Incident Count = [0, 7.8]	0.097	0.675	3.588
Summer	Incident Count = (7.8, 15.6]	0.090	0.343	0.906
Spring	Incident Count = (7.8, 15.6]	0.090	0.343	0.906

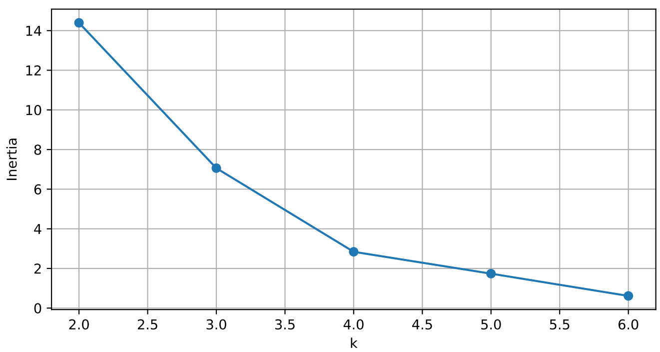

Fig. 24. Inertia vs k values of district data set.

VII. CLUSTERING

In order to analyze the data set from an unsupervised perspective, several clustering algorithms were performed in this section. These methods can be grouped into three main categories: partitioning based, hierarchical and density based clustering.

The original data set consists of incident observations, each row corresponding to an individual event. It does not permit districts or streets to be treated as comparable units for clustering. Clustering methods require each object to be represented as a single point in a common feature space. In order to employ clustering, two data sets were prepared. The first data set was a district representation where each entry corresponds to a district and was described by aggregated incident indicators. Districts with insufficient incident volumes were excluded. Specifically, Balçova and Narlıdere were not included since their total incident counts did not exceed 200. The second data set was a street representation, focusing on incident-dense streets and consisting of the top 15 streets with the highest incident counts. For both data sets, the indicators were Average Intervention Time, Monthly Average Incidents, Number of Life-threatening Accidents, and Number of Non-life-threatening Accidents happened in each district or street.

Before applying clustering algorithms, the Hopkins statistic was computed to examine whether the two representations have a meaningful clustering tendency. For the district data set, the Hopkins value was 0.542 which indicated only a moderate deviation from randomness. For the street data set, the Hopkins statistic was higher, 0.686, implying a greater tendency toward clusterable structure.

Next, a partitioning approach was adopted by using the k-means algorithm. For each data set, multiple of k were evaluated using two metrics: the silhouette score and inertia. The

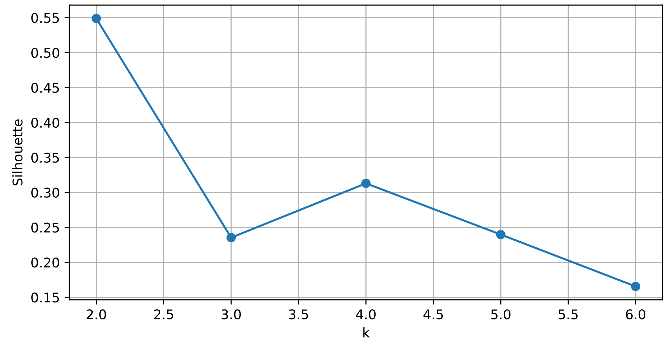

Fig. 25. Silhouette score vs k values of district data set.

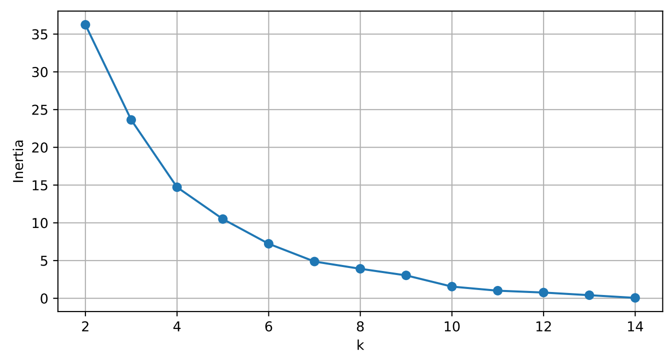

Fig. 26. Inertia vs k values of street data set.

corresponding silhouette (Figure 25- 27) and inertia curves (Figure 24- 26) were provided for both the district and street-based representations. Considering the elbow behaviour in the inertia plots, the final selections were k as 4 for the district representation and k as 6 for the street representation. The analysis was carried out for cluster sizes, centroid behaviors, and visualizations.

For the district based representation, four clusters were identified. The cluster size distribution was $3, 1, 2, 2$ for clusters $C0$–$C3$, respectively. The corresponding cluster centroids were summarized and visualized as centroid profiles in Figure 28 and Figure 30. Moreover, a two-dimensional

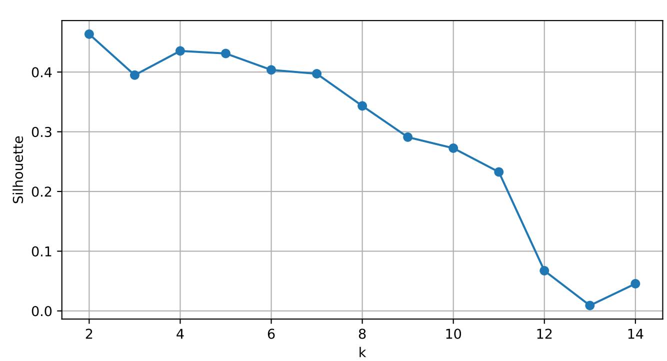

Fig. 27. Silhouette score vs k values of street data set.

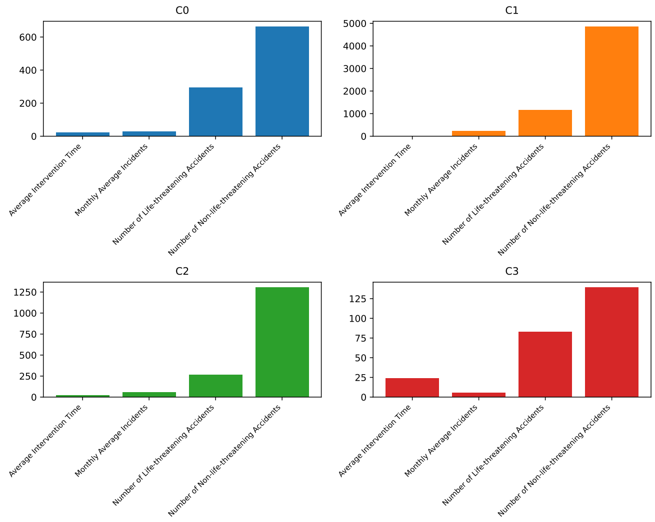

Fig. 28. Shape of cluster centers of district dataset by K-means algorithm

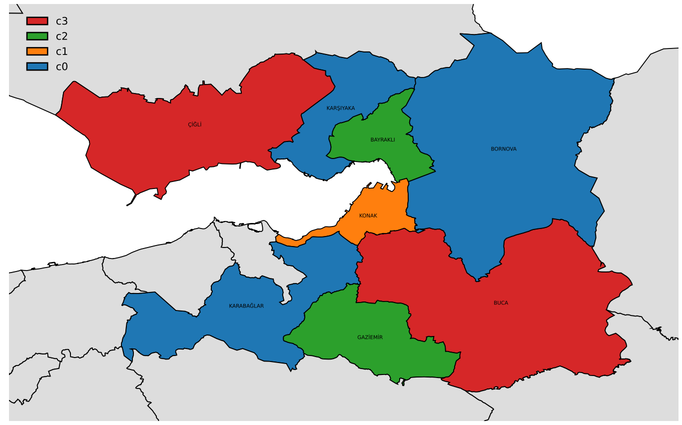

Fig. 30. The map of districts in Izmir based on the cluster labels.

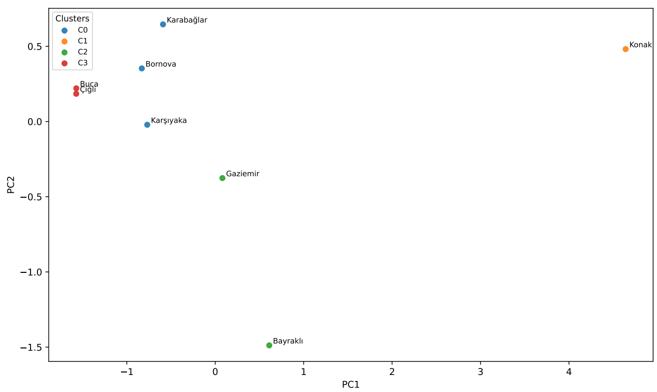

Fig. 29. PCA projection of districts into the first two principal components, coloured by k-means cluster membership.

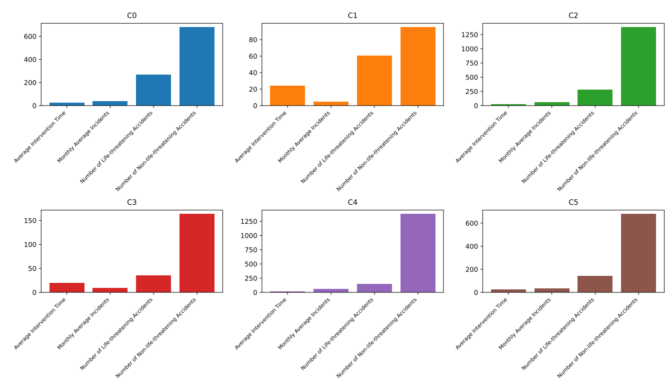

Fig. 31. Shape of cluster centers of street datasetby K-means algorithm

PCA projection coloured by resulted clusters was provided in Figure 29.

For the street base representation, six clusters were identified. The resulting cluster sizes were $2, 5, 3, 2, 1, 2$ for clusters $C0$–$C5$, respectively. The cluster centroid profiles were shown in Figure 31 as a summary. Also, the cluster structure was visualized using a PCA scatter plot in Figure 32, where streets were coloured according to their assigned cluster.

In addition to the k-means algorithm, the k-median algorithm was implemented and evaluated with the same data sets for comparison. The results can be found in the GitHub repository[6].

Revisiting the overall incident count distribution presented in Figure 2 provides the necessary context for the clustering strategy. In this figure, Konak substantially dominates the other districts in terms of total incident count, while the remaining districts exhibit significantly lower numbers, forming a sec-

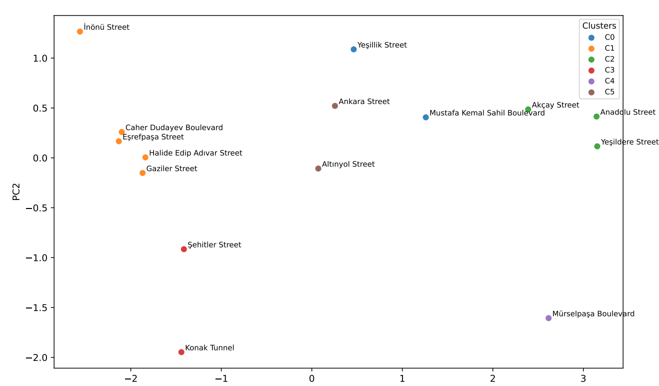

Fig. 32. PCA projection of incident-dense streets into the first two principal components, coloured by k-means cluster membership

[6]https://github.com/HuseyinYontar/Analyzing_Traffic_Incidents_in_Izmir/tree/main/clustering/clustering_for_report/kmedians_outputs

ondary tier. This pattern is consistent with the district-based K-means clustering, where one cluster (C1; size=1) emerges as a distinct outlier characterized by the highest incident intensity (monthly average incidents = 224.61) and the largest life-threatening and non-life-threatening totals (1,155 life-threatening; 4,849 non-life-threatening). Apart from the $C1$ cluster, $C2$ (size=2) captures high incident districts (monthly average = 56.38), $C0$ (size=3) represents moderate incident districts (monthly average = 29.83), and $C3$ (size=2) corresponds to low incident districts (monthly average = 5.70). Furthermore, average intervention times vary significantly across clusters. $C1$ has the shortest mean intervention time (19.12), whereas other clusters exhibit longer mean times ($C0$: $\approx$23, $C2$: $\approx$20.25, $C3$: 23.5). This observation suggests that areas with the greatest demand may benefit from better operational accessibility.

When the street based clustering was considered, the clusters appear to be primarily formed by their severity. However, the top four streets with highest incidents were split across two clusters due to multidimensional differences in their profiles. $C4$ (Mürselpaşa Bulvarı) was separated from the $C2$ (other top streets) because its average intervention time was substantially lower at 16.18 minutes, compared to approximately 20 minutes in cluster $C2$. When clusters $C0$ and $C5$ were examined, $C0$ revealed an interesting pattern: the two streets differed substantially in monthly average incidents (Mustafa Kemal Sahil Boulevard: 41.78 vs. Yeşillik Street: 27.76), yet their life-threatening incident totals were relatively similar (253 vs. 280). This suggests that $C0$ groups streets that may have different incident counts but show similar severity patterns, so grouping them together makes sense from a risk management perspective. In contrast, $C5$ represents a similar mean incident profile (Ankara Street: 34.93, Altınyol Street: 30.17) with similar mean intervention times, supporting the argument that the clusters were structured by incident density. For the remaining clusters, grouping was driven primarily by monthly average incident rates. $C3$ contains relatively higher low-incident streets (Şehitler Street: 9.50, Konak Tunnel: 7.29), while $C1$ groups the lowest intensity corridors with monthly average incidents clustered around 3.17–5.42.

In addition to the partitioning-based clustering, hierarchical clustering was performed. For this task, an agglomerative (bottom-up) strategy was adopted where observations were progressively merged according to their distances to produce a dendrogram. To further investigate the hierarchical structure, two linkage criteria were considered: single linkage and complete linkage.

For both the district and street based representations, dendrograms were generated under each linkage criterion and reported in Figures 33–34 and Figures 35–36, respectively. These visualizations provided a summary of how points merge into larger groups as the linkage distance increases, comparing the behaviors of complete linkage versus single linkage across the two data sets.

Both linkage methods show that Konak stands apart as a distinct outlier in the district dendrograms, connecting to the other districts only at the maximum dissimilarity threshold. In contrast, the remaining districts cluster together at considerably lower levels, creating a fairly unified group with some closely paired subsets.

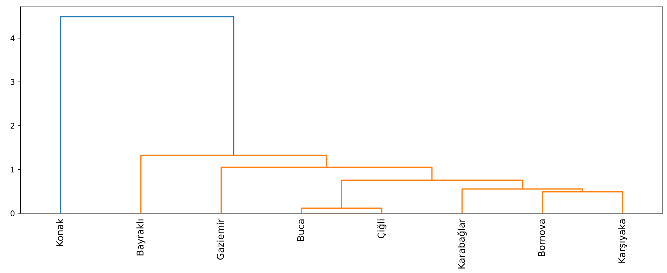

Fig. 33. Hierarchical clustering dendrogram for the district representation using single linkage.

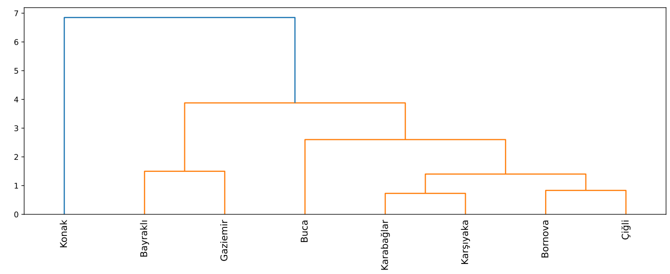

Fig. 34. Hierarchical clustering dendrogram for the district representation using complete linkage.

The street-level hierarchies generally produce a structure aligned with the K-means solution. Streets with high incident number form a dedicated branch, while streets with lower incident numbers subdivide into small groups with similar average incident numbers. As expected, single linkage shows progressive aggregation through nearest neighbors, whereas complete linkage yields more clearly separated groups, making the high-moderate–low incident density separation visually more explicit.

As a final clustering method, a density based clustering approach was considered using DBSCAN. The behaviour of DBSCAN was controlled by two parameters: the neighborhood radius ε and the minimum number of points required to form

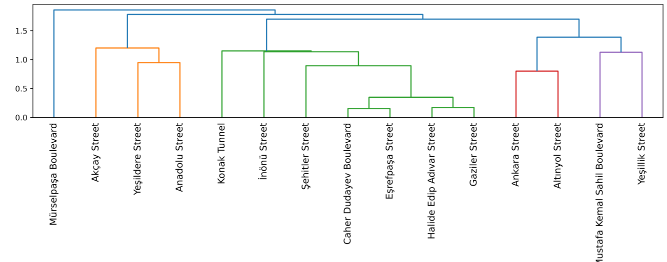

Fig. 35. Hierarchical clustering dendrogram for the street representation using single linkage.

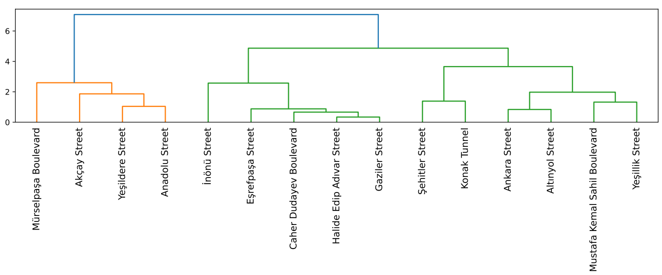

Fig. 36. Hierarchical clustering dendrogram for the street representation using complete linkage.

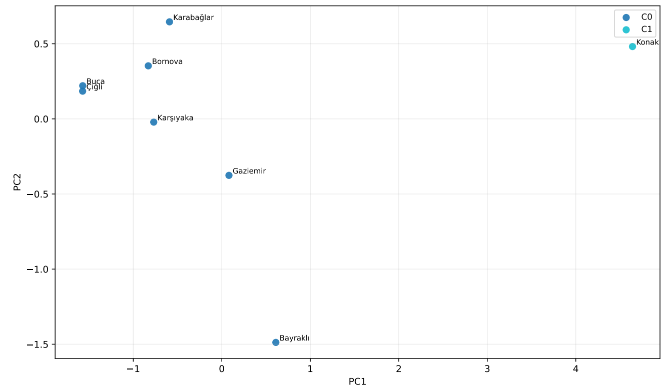

Fig. 37. PCA projection of the district representation, coloured by DBSCAN cluster labels.

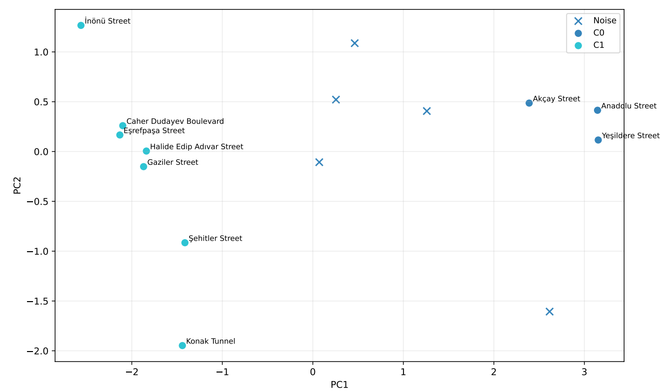

Fig. 38. PCA projection of incident-dense streets, coloured by DBSCAN cluster labels; noise points are marked separately.

a dense region. To determine an appropriate configuration, a grid search over the $(\varepsilon, minsamples)$ space was performed. The results were evaluated using the silhouette score which was computed after removing the noise points.

For the district representation, the best-performing configuration in the grid search was $minsamples = 1$ and $\varepsilon = 1.4$, yielding 2 clusters ($C0$ with size 7, $C1$ with size 1) with 0 noise points and a silhouette score of 0.648. The resulting cluster structure was visualized using a PCA projection illustrated in Figure 37. As shown in the figure, DBSCAN produces a very clear split: Konak forms its own cluster ($C1$), while all other districts fall into a single cluster ($C0$). In the PCA space, Konak was separated mainly along PC1, suggesting its incident profile is fundamentally different from the rest.

For the street representation, the highest silhouette score of 0.735 (noise removed) was obtained with $minsamples = 3$ and $\varepsilon = 1.2$, producing 2 clusters ($C0$ with size 7 and $C1$ with size 3) and 5 noise points. PCA visualization of DBSCAN was provided in Figure 38. It can be observed from the figure that $C0$ was formed by the streets having the highest monthly incident average counts, while $C1$ groups the streets with lower monthly incident average counts. When the noise points were considered: the right lower point was Mürselpaşa Boulevard which has high number of incidents and lower average intervention time compared to the $C0$, the middle points Mustafa Kemal Sahil Boulevard, Ankara Street, Altınyol Street and Yeşillik Street have intermediate monthly average incident counts ($\approx$34), which was not close to either $C0$ ($\approx$59.5) or the $C1$ ($\approx$5.5).

VIII. Ensemble Learning

In this section, ensemble learning methods were employed to predict whether an incident was life-threatening. The training and test sets utilized were identical to the data sets used in the previous classification section. The two ensemble learning algorithms selected for this task were Random Forest and AdaBoost, both of which were widely recognized in the field.

Table VI presents the performance assessment of the trained ensemble learning models by summarizing overall accuracy, F1-score, class-wise precision and recall, as well as the confusion matrix components (TN, FP, FN, TP).

To determine the optimal combination of hyperparameters for both models, the Grid Search method was applied. Both models achieved their optimal performance using 400 base estimators.

For Random Forest, the best configuration yielded an accuracy of approximately 0.72 and an F1-score of approximately 0.70 for the life-threatening class. In comparison, for the AdaBoost model, the best configuration yielded an accuracy of 0.71 and an F1-score of approximately 0.67 for the life-threatening class.

Overall, neither ensemble method surpassed the neural network classifier reported in the previous classification / regression section.

IX. Conclusion

This study provides a comprehensive data-driven analysis of traffic incident patterns in Izmir's metropolitan area, examining 21,161 incident records spanning December 2021 to September 2025. Through the application of multiple analytical methodologies including classification, time-series regression, association mining, clustering, and ensemble learning, the research identified critical spatial and temporal patterns that can inform evidence-based traffic safety interventions.

TABLE VI
DETAILED PERFORMANCE METRICS OF THE ENSEMBLE MODELS

Model	Accuracy	Precision		Recall		F1-score		Confusion Matrix			
		Life-threatening	Non-life-threatening	Life-threatening	Non-life-threatening	Life-threatening	Non-life-threatening	TN	FP	FN	TP
Random Forest	0.7212	0.75	0.70	0.66	0.79	0.70	0.74	485	132	212	405
AdaBoost	0.7066	0.77	0.67	0.59	0.82	0.67	0.74	508	109	253	364

The spatial analysis revealed a striking concentration of incidents in Konak district, establishing it as priority district for traffic safety interventions. The analysis further identified specific infrastructure hotspots that are bridge overpasses on Mürselpaşa Boulevard and Yeşildere Avenue. This spatial clustering pattern aligned with Haybat and Karakaş's [2] findings, which documented incident concentration in high-mobility regions. The temporal analysis revealed distinct patterns between working and non-working days. Weekdays followed the characteristic double-peak pattern at 08:00 and 17:00, aligning with Cardelino's [4] documentation of rush-hour traffic volume distributions. Non-working days displayed a flatter profile with a single mid-afternoon peak around 16:00, suggesting different trip purposes compared to structured weekday commutes. The most critical finding concerned incident severity by time of day. Late-night hours experienced substantially lower incident volumes but dramatically elevated injury and fatal crash proportions, reaching 88% during early morning hours compared to 22-30% during daytime peaks. This pattern indicates nighttime crashes, though less frequent, were substantially more likely to result in life-threatening outcomes. Classification models for predicting incident severity achieved encouraging results, with MLP neural network attaining 73.82% accuracy. Logistic Regression, Random Forest, and Decision Tree achieved approximately 70% accuracy. Decision Tree analysis identified late-nighttime intervals (20:00-07:54) as the most critical incident severity predictor. Ensemble methods including Random Forest and AdaBoost showed promising results though did not surpass MLP's performance. Regression analysis revealed substantial location-specific variation in forecasting performance, demonstrating distinct temporal dynamics across areas. SARIMA excelled for district-level predictions, MLP for specific infrastructure locations, and ARIMA for other streets. These variations indicate tailored forecasting strategies outperform uniform approaches, requiring customized models for different spatial contexts.

Unsupervised clustering identified distinct risk profiles across districts and streets based on incident volume, severity, and intervention times. K-means separated districts into four groups, with Konak emerging as a unique high-intensity outlier, corroborated by DBSCAN (silhouette score: 0.648). Street-level clustering revealed six groups differentiated by frequency and operational characteristics. Ensemble methods validated findings, with Random Forest and AdaBoost achieving 72% and 71% accuracy respectively, though not surpassing MLP. Association mining yielded limited insights beyond confirming Sundays' low incident count pattern. These analyses confirm that distinct, location-based risk profiles require tailored intervention strategies.

This study demonstrated that systematic, data-driven analysis of municipal traffic incident records yielded actionable insights for improving urban traffic safety. By targeting the identified high-risk locations, times, and incident characteristics, municipal authorities can implement targeted interventions to maximize safety improvements in Izmir and other metropolitan areas facing similar challenges.

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